a **The molecular mechanism, as the models take it**

AD **Alzheimer's disease**

The Aβ42 pool is set by clearance, so a miR-29 mimic acts on the production side, through BACE1

1 Repression

BACE1 mRNA

3′UTR

miR-29

t½ 7 and 10.6 h

AGO2 · 6N4O

mimic

2 Production

BACE1 t½ ≈19 h

BACE1 · 4D8C

APP cut at the β site

Aβ42

3 Downstream

1.41

production 6.6 %/h

clearance 5.3 %/h

controls 6.7 and 7.6

Aβ42 · 5OQV

PD **Parkinson's disease**

α-synuclein sits on the rising part of its elongation curve, so a knockdown reaches the fibril less than 1:1

1 Repression

SNCA mRNA

3′UTR

miR-7

t½ 1.7 h

AGO2 · 6N4O

mimic

2 Production

21.6 μM in a bouton

half-life ≥ 24 h

α-synuclein monomer

3 Downstream

rate

0.5

0.0

0

60

120 μM

30–32% of the maximum rate

−30% protein: −22.6–23.0% rate

α-syn · 6CU7

b **How the mechanistic algorithm runs**

1 Read the parameter table

measured

67 rows from 15 primary sources

54 measured, 4 derived, 8 qualitative

1 declared gap: stops the run

A row loads only if measured, or derived and requested by name.

2 Closed form at steady state

scanned

mimic factor f

100

10

1

0.01

1

100

ρ

κ = 1 − 1/R, from the table alone

f = 1 + κ/(1−κ) · (1+ρ)/ρ, ρ scanned

3 Follow the repeated dose

assumed

E

μ

π

bolus D every T h on the repressor E, then mRNA μ, protein π (and Aβ42 a). The mimic is cleared like the endogenous species.

4 Solve for the dose

computed

peak of E

10^4

100

1

8

24

168

interval T (h)

One cycle is an affine map; solve for D so that the cycle mean of the protein falls by κ.

5 Sweep ρ and T, then rank

computed

T (h)

168

24

2

0.01

1

100

ρ

T within 10×

5 h

53 h

1.7 h

20 h

49 values of ρ × 19 intervals × 9 series; rank by the longest T within a 3×, 10× or 30× peak.

6 Check every step

computed

closed form against the ODE: 3×10^{-10}

six independent calculations of the dosing code, worst 0.056 of tolerance

grid against the scenarios at 24 h, 63 points: 6×10^{-10}

consistency checks between tables, models and figures (scripts/08)

A step rests on: measured: from the source sentences — scanned: unmeasured, swept over a range ····· assumed: stated

Renders: deposited structures, as illustration; no number is read from them. Small plots: drawn from the result tables of Figs. 3 to 8.